Reproducing Table I of “Tracking down Quirks at the LHC”

Knapen, Lou, Papucci & Setford, arXiv:1708.02243 · paper-stated acceptance with an inferred longitudinal detector model · 3 October 2026

Setup

- **Stated in the paper or its detector reference [27]:** pixel and SCT barrel only ($|\eta| \lesssim 1.8$); pixel-centre hits; neighbouring pixels merged; SCT 34 μm across (2 × 17 μm); 45 mm beam spot; 25 ns timing; trigger pT(QQ) > 200 GeV; the paper's plane-finding algorithm and signal region.
- **Our reading (not stated):** SCT smearing along the beam of 1.16 mm (2 × the 0.58 mm in [27]); the 45 mm beam spot as one common shift per event, with the fit measured from the true vertex.
- **Also shown:** 0.58 mm (the value in [27], not doubled) and no smearing along the beam (no source).
- Two independent samples per benchmark (1000 and 2000 generated events), never combined. No tuning to the paper.

Definitions

ε_{trig} = N(pT of the quirk pair > 200 GeV) / N(matched events)
ε_{fid} = N(triggered, every quirk crosses all 8 layers inside $|\eta| < 1.8$, within 25 ns) / N(triggered)
ε_{reco} = N(fiducial and plane found, 2 hits in 7 layers + ≥ 1 in the 8th, $\Delta w < 1$ cm, $\Delta s < 0.01$ cm) / N(fiducial)
$\varepsilon = \varepsilon_{trig} \times \varepsilon_{fid} \times \varepsilon_{reco}$ (paper Eq. 12)

Colored only: both R-hadrons must be charged; this cut is applied to the triggered events before ε_{fid} (27 of 101 and 68 of 207 events). The paper does not say which factor contains it.

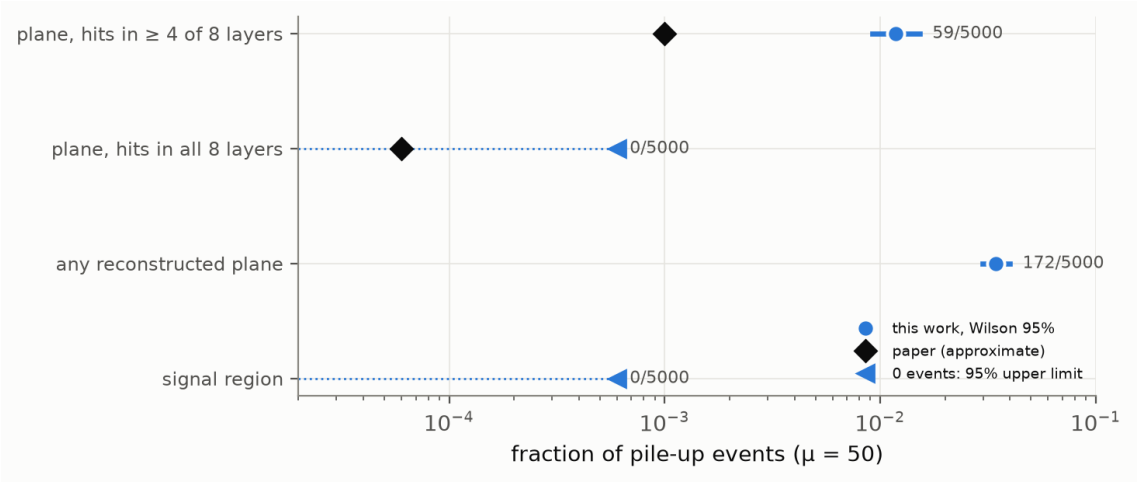
Standard error: $SE = \sqrt{\varepsilon (1 - \varepsilon) / n}$
Wilson 95% interval: $[\ \varepsilon + z^2/2n \pm z \sqrt{\varepsilon(1-\varepsilon)/n + z^2/4n^2}\ ] / (1 + z^2/n)$, $z = 1.96$
Upper limit (UL): when 0 of n events pass, the largest true rate that still gives 0 events in at least 5% of experiments: $UL = 1 - 0.05^{1/n} \approx 3/n$
p-value (exact two-sided binomial test): if the paper's value were exactly right, the probability of getting a count as unlikely as ours or more (the paper gives no counts or errors, so its own uncertainty is ignored)

p-value	Verdict
$p < 0.01$	**Strongly inconsistent**
$0.01 \le p < 0.05$	**Statistically inconsistent**
$0.05 \le p < 0.10$	**Borderline / mild tension**
$p \ge 0.10$	**No statistically significant inconsistency**

1. Pile-up background (⟨μ⟩ = 50, 5000 events)

Quantity	k/n	Rate ± SE	Wilson 95% / UL	Paper	Comparison
Plane, hits in ≥ 4 of 8 layers	59/5000	$1.2\times10^{-2} \pm 1.5\times10^{-3}$	$[9.2\times10^{-3}, 1.5\times10^{-2}]$	$\approx 1\times10^{-3}$	Higher than the paper's approximate rate
Plane, hits in all 8 layers	0/5000	0	UL 6.0×10^{-4}	$\approx 6\times10^{-5}$	Compatible with the paper's approximate rate
Any reconstructed plane	172/5000	$3.4\times10^{-2} \pm 2.6\times10^{-3}$	$[3.0\times10^{-2}, 4.0\times10^{-2}]$	—	—
Hit multiplicity	0/5000	0	UL 6.0×10^{-4}	—	No event passes
Signal region	0/5000	0	UL 6.0×10^{-4}	—	No event passes

The paper gives approximate rates without counts or errors, so no formal test is made.



Pile-up rates with Wilson 95% intervals; ◁ = 95% upper limit when no event passes; diamonds = paper.

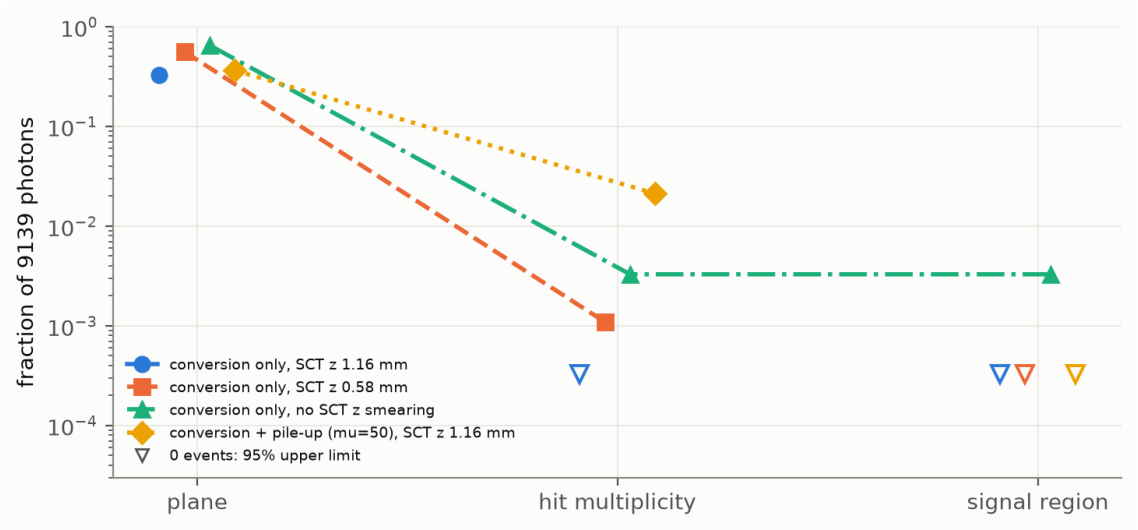
Why the 4-of-8 rate is higher

- Our pile-up tracks are ideal helices in the 8 barrel layers; the paper's simulation adds energy loss, bremsstrahlung and endcaps, which remove soft tracks.
- Our minimum-bias pool has 20 000 events (paper: 125 000), and the paper does not define “a plane” for this number.

2. Converted-photon background (9139 Z(νν)γ+jet photons, |η| < 1.8, fit from the vertex)

Setup	Plane	Hit mult.	Signal region	Wilson 95% / UL	Events at 300 fb^{-1} (our σ / paper σ)
conversion only SCT z 1.16 mm	3005/9139	0/9139	0/9139	UL 3.3×10^{-4}	< 0.21 / < 0.98
conversion only SCT z 0.58 mm	5124/9139	10/9139	0/9139	UL 3.3×10^{-4}	< 0.21 / < 0.98
conversion only no SCT z smearing	5937/9139	30/9139	30/9139	$[2.3\times10^{-3}$, $4.7\times10^{-3}]$	2.1 / 9.8
conversion + pile-up (μ = 50) SCT z 1.16 mm	3323/9139	193/9139	0/9139	UL 3.3×10^{-4}	< 0.21 / < 0.98

Events at 300 fb^{-1} = σ × 1% conversion probability × fraction × 300 fb^{-1}, with σ = 0.21 pb (our MadGraph) or ≈ 1 pb (paper). With 0 events the 95% upper limit is used.



Fraction of photons passing each cut; open triangles = 95% upper limit when no event passes.

Reading

- With SCT smearing along the beam (1.16 or 0.58 mm), no conversion passes, with or without pile-up, as the paper says.
- Pile-up hits can fill the gaps of an $e^{+}e^{-}$ pair (193/9139 pass the hit count), but none of them is thin enough to pass Δs.
- Without SCT smearing along the beam, 30/9139 = 0.33% of conversions pass. That setting is not background-free.
- Our cross-section is 4.3× below the paper's ≈ 1 pb (generator settings not given). The 1% conversion probability is taken from the paper's own "≈ 1 pb → ≈ 10 fb".

3. Colored quirks (m_Q = 1800 GeV, Λ = 2 keV)

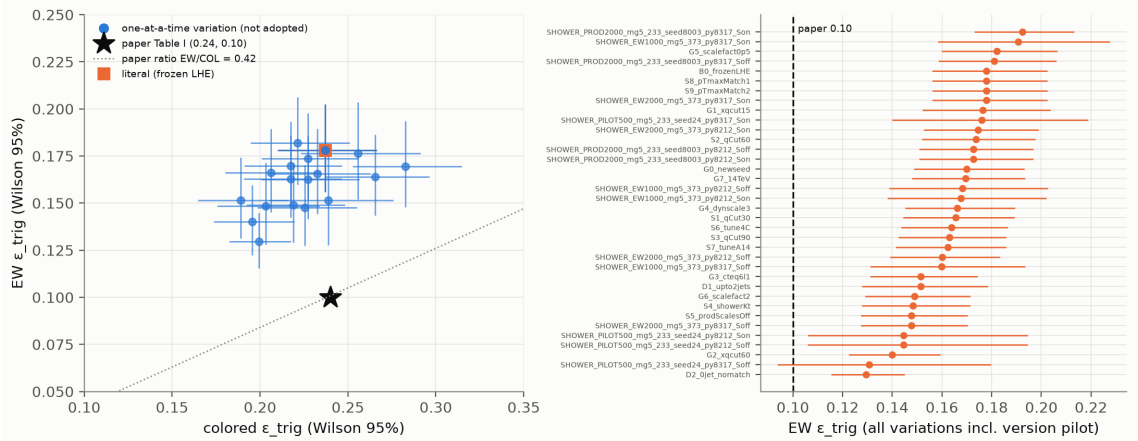
Efficiency	Sample	k/n	ε ± SE	Wilson 95%	Paper	p	Verdict
ε_{trig}	COL1000	101/419	0.241 ± 0.021	[0.203, 0.284]	0.24	0.95	No statistically significant inconsistency
ε_{trig}	COL2000	207/873	0.237 ± 0.014	[0.210, 0.266]	0.24	0.87	No statistically significant inconsistency
ε_{fid}	COL1000	7/27	0.259 ± 0.084	[0.132, 0.447]	0.28	1.00	No statistically significant inconsistency
ε_{fid}	COL2000	25/68	0.368 ± 0.058	[0.263, 0.486]	0.28	0.14	No statistically significant inconsistency
ε_{reco}, SCT z 1.16 mm	COL1000	1/7	0.143 ± 0.132	[0.026, 0.513]	0.35	0.43	No statistically significant inconsistency
ε_{reco}, SCT z 1.16 mm	COL2000	3/25	0.120 ± 0.065	[0.042, 0.300]	0.35	0.019	Statistically inconsistent
ε_{reco}, SCT z 0.58 mm	COL1000	3/7	0.429 ± 0.187	[0.158, 0.750]	0.35	0.70	No statistically significant inconsistency
ε_{reco}, SCT z 0.58 mm	COL2000	5/25	0.200 ± 0.080	[0.089, 0.391]	0.35	0.14	No statistically significant inconsistency
ε_{reco}, SCT z none	COL1000	5/7	0.714 ± 0.171	[0.359, 0.918]	0.35	0.056	Borderline / mild tension
ε_{reco}, SCT z none	COL2000	11/25	0.440 ± 0.099	[0.267, 0.629]	0.35	0.40	No statistically significant inconsistency

4. Electroweak quirks (m_Q = 800 GeV, Λ = 4 keV)

Efficiency	Sample	k/n	ε ± SE	Wilson 95%	Paper	p	Verdict
ε_{trig}	EW1000	89/511	0.174 ± 0.017	[0.144, 0.209]	0.10	2.8×10^{-7}	**Strongly inconsistent**
ε_{trig}	EW2000	186/1045	0.178 ± 0.012	[0.156, 0.202]	0.10	1.8×10^{-14}	**Strongly inconsistent**
ε_{fid}	EW1000	40/89	0.449 ± 0.053	[0.350, 0.553]	0.28	8.2×10^{-4}	**Strongly inconsistent**
ε_{fid}	EW2000	94/186	0.505 ± 0.037	[0.434, 0.576]	0.28	1.3×10^{-10}	**Strongly inconsistent**
ε_{reco}, SCT z 1.16 mm	EW1000	6/40	0.150 ± 0.056	[0.071, 0.291]	0.72	9.1×10^{-14}	**Strongly inconsistent**
ε_{reco}, SCT z 1.16 mm	EW2000	21/94	0.223 ± 0.043	[0.151, 0.318]	0.72	2.4×10^{-23}	**Strongly inconsistent**
ε_{reco}, SCT z 0.58 mm	EW1000	10/40	0.250 ± 0.068	[0.142, 0.402]	0.72	9.4×10^{-10}	**Strongly inconsistent**
ε_{reco}, SCT z 0.58 mm	EW2000	29/94	0.309 ± 0.048	[0.224, 0.408]	0.72	1.5×10^{-16}	**Strongly inconsistent**
ε_{reco}, SCT z none	EW1000	34/40	0.850 ± 0.056	[0.709, 0.929]	0.72	0.078	**Borderline / mild tension**
ε_{reco}, SCT z none	EW2000	76/94	0.809 ± 0.041	[0.717, 0.875]	0.72	0.065	**Borderline / mild tension**

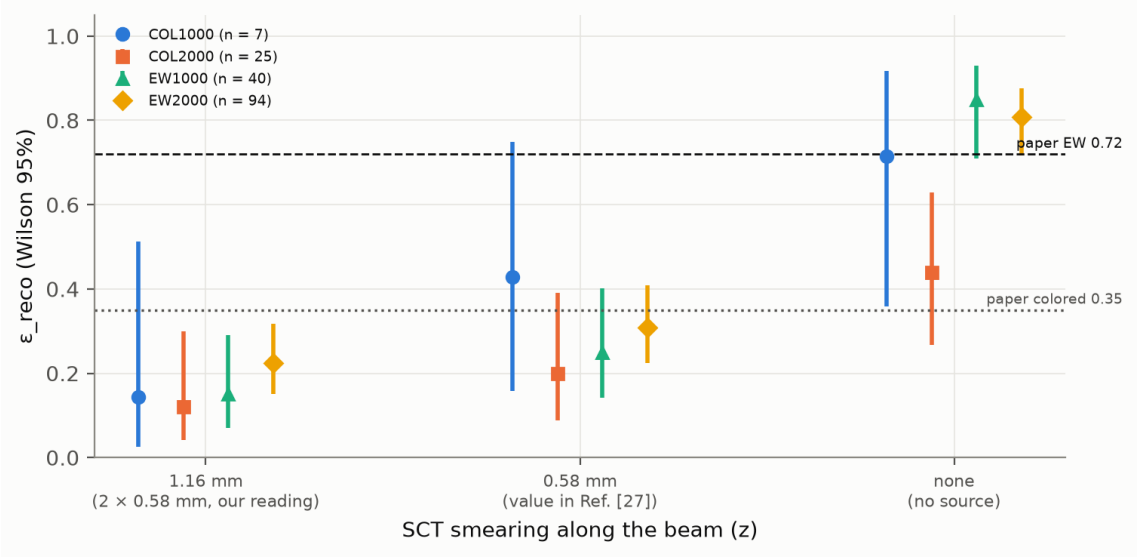
Reading

- **Colored:** trigger and fiducial agree with the paper. Reconstruction agrees for 0.58 mm and for no smearing; at 1.16 mm the larger sample is low (3/25).
- **EW trigger:** 17–18% against 10%. The same definition gives the paper's colored value, and 36 generator variations keep EW at 13–19% (figure below). The paper's generator settings are not given.
- **EW fiducial:** 45–51% against 28%. Using the detector lengths of [27] instead of $|\eta| < 1.8$ gives 33–37%, close to 28% (section 6), but the paper does not say it used them.
- **EW reconstruction:** 15–31% with SCT smearing along the beam; 81–85% only without it.

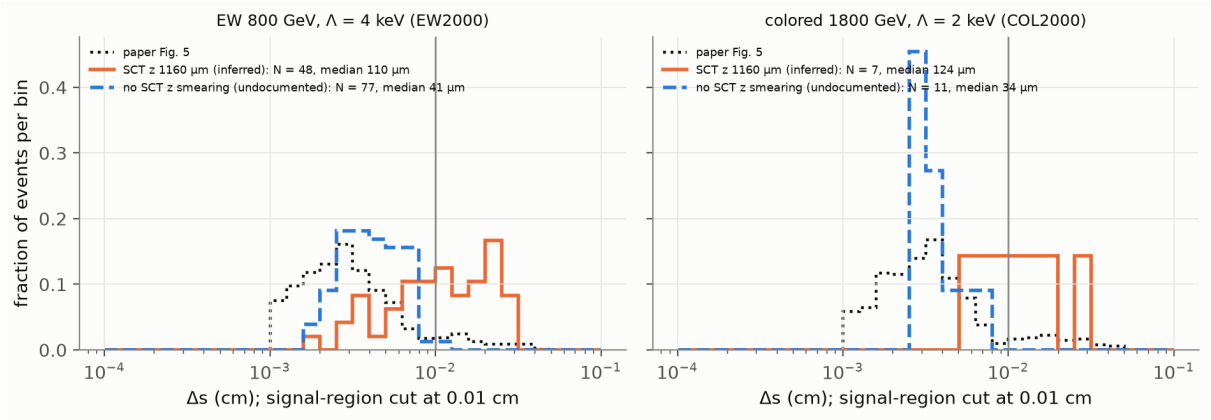


EW vs colored trigger efficiency for every generator variation tried (left; star = paper) and all EW values (right).

5. What decides reconstruction



Reconstruction efficiency of the same events for each SCT smearing along the beam. Bars = Wilson 95%; lines = paper.



Signal Δs before the last cut. With 1.16 mm (solid) our events sit right of the paper's Fig. 5 (dotted); without smearing (dashed) they move close to it.

Reading

- [27] gives the SCT accuracy as 17 µm across and 0.58 mm along the beam. The paper doubles “the intrinsic accuracy” without naming the direction, so 1.16 mm is our reading of the text.
- With 1.16 mm the reconstructed plane is too thick (signal Δs median 110–124 µm; cut 100 µm). The paper's own Fig. 5 (≈ 30 µm) fits only with little or no smearing along the beam.

45 mm beam spot. The paper smears “the z-coordinates of all the hits” and assumes the vertex “is identified correctly, and is located at the origin”. Events passing the signal region for SCT z 1.16 · 0.58 mm · none:

Reading of the beam-spot sentence	Colored 2000	EW 2000
One common shift per event, fit from the true vertex (used)	3 · 5 · 11 of 25	21 · 29 · 76 of 94
One common shift per event, fit from the origin	0 · 0 · 0 of 25	3 · 3 · 4 of 94
Separate 45 mm shift of every hit	0 · 0 · 0 of 21	0 · 0 · 0 of 66

6. Table I: paper vs this work

	Quantity	Paper	1000-event sample: ε ± SE, Wilson 95%, p	2000-event sample: ε ± SE, Wilson 95%, p	Verdict (2000)
Colored	ε_{trig}	0.24	101/419 = 0.241 ± 0.021 [0.203, 0.284], p = 0.95	207/873 = 0.237 ± 0.014 [0.210, 0.266], p = 0.87	**No statistically significant inconsistency**
	ε_{fid}	0.28	7/27 = 0.259 ± 0.084 [0.132, 0.447], p = 1.00	25/68 = 0.368 ± 0.058 [0.263, 0.486], p = 0.14	**No statistically significant inconsistency**
	ε_{reco}, z 1.16 mm	0.35	1/7 = 0.143 ± 0.132 [0.026, 0.513], p = 0.43	3/25 = 0.120 ± 0.065 [0.042, 0.300], p = 0.019	**Statistically inconsistent**
	ε_{reco}, z 0.58 mm	0.35	3/7 = 0.429 ± 0.187 [0.158, 0.750], p = 0.70	5/25 = 0.200 ± 0.080 [0.089, 0.391], p = 0.14	**No statistically significant inconsistency**
	ε_{reco}, no z smear	0.35	5/7 = 0.714 ± 0.171 [0.359, 0.918], p = 0.056	11/25 = 0.440 ± 0.099 [0.267, 0.629], p = 0.40	**No statistically significant inconsistency**
EW	ε_{trig}	0.10	89/511 = 0.174 ± 0.017 [0.144, 0.209], p = 2.8×10^{-7}	186/1045 = 0.178 ± 0.012 [0.156, 0.202], p = 1.8×10^{-14}	**Strongly inconsistent**
	ε_{fid}	0.28	40/89 = 0.449 ± 0.053 [0.350, 0.553], p = 8.2×10^{-4}	94/186 = 0.505 ± 0.037 [0.434, 0.576], p = 1.3×10^{-10}	**Strongly inconsistent**
	ε_{reco}, z 1.16 mm	0.72	6/40 = 0.150 ± 0.056 [0.071, 0.291], p = 9.1×10^{-14}	21/94 = 0.223 ± 0.043 [0.151, 0.318], p = 2.4×10^{-23}	**Strongly inconsistent**
	ε_{reco}, z 0.58 mm	0.72	10/40 = 0.250 ± 0.068 [0.142, 0.402], p = 9.4×10^{-10}	29/94 = 0.309 ± 0.048 [0.224, 0.408], p = 1.5×10^{-16}	**Strongly inconsistent**
	ε_{reco}, no z smear	0.72	34/40 = 0.850 ± 0.056 [0.709, 0.929], p = 0.078	76/94 = 0.809 ± 0.041 [0.717, 0.875], p = 0.065	**Borderline / mild tension**

Bottom line

- **Agrees with the paper:** colored trigger and fiducial efficiency; colored reconstruction for 0.58 mm or no SCT smearing along the beam.
- **Does not agree:** EW trigger (0.18 vs 0.10), EW fiducial (0.51 vs 0.28), and EW reconstruction with SCT smearing along the beam (0.22–0.31 vs 0.72).
- **Backgrounds:** pile-up only and conversion only give no signal-region event, and so does the conversion + pile-up overlay with this setup. Without SCT smearing along the beam, 0.33% of conversions would pass.
- **Why a full reproduction is not possible:** the paper does not fix the SCT smearing along the beam, the beam-spot treatment, the exact acceptance or the generator settings, and it gives no event counts or errors.

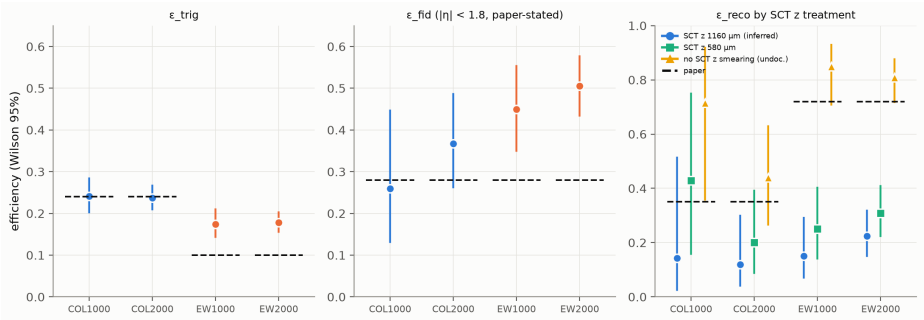


Table I comparison; reconstruction shown for each SCT smearing along the beam. Bars = Wilson 95%; dashed = paper.

Sensitivity check: detector lengths from [27] (pixel |z| < 40.05 cm, SCT |z| < 74.9 cm) instead of |η| < 1.8

Sample	ε_fid = k/n ± SE	Wilson 95%	p vs 0.28	Signal region (z 1.16 · 0.58 mm · none)
COL1000	6/27 = 0.222 ± 0.080	[0.106, 0.408]	0.67	1 · 2 · 5 of 6
COL2000	22/68 = 0.324 ± 0.057	[0.224, 0.442]	0.42	1 · 3 · 8 of 22
EW1000	33/89 = 0.371 ± 0.051	[0.278, 0.475]	0.059	2 · 5 · 27 of 33
EW2000	62/186 = 0.333 ± 0.035	[0.270, 0.404]	0.12	10 · 14 · 50 of 62

Appendix A. Where every setting comes from

Green = stated in the paper or [27]; amber = our reading; red = no source. Counts are listed in the order COL1000, COL2000, EW1000, EW2000.

Setting	Source	Status	Resulting counts
Barrel only (pixel + SCT, 8 layers)	Paper Sec. III A: 'we only included the barrel of the pixel and SCT detectors in our simulations'	**PAPER-STATED**	frame for all counts
Acceptance: effective |eta| < 1.8, per hit, no finite-z boundary	Paper Sec. III A: '... which effectively restricts the fiducial range to |eta| <~ 1.8' (applying it as a per-hit cut is the implementation)	**PAPER-STATED**	fiducial COL1000 7/27, COL2000 25/68, EW1000 40/89, EW2000 94/186
Acceptance: finite barrel lengths (pixel |z|<400.5 mm, SCT |z|<749 mm, IBL |z|<330.15 mm)	Ref. [27] (arXiv:0901.0512 pp. 40-43) table 'Sensitive barrel 0 < |z| < 400.5 / 749'; IBL arXiv:1803.00844. No evidence the authors applied them	**INFERRED (sensitivity)**	fiducial COL1000 6/27, COL2000 22/68, EW1000 33/89, EW2000 62/186
Layer radii IBL 33.25, pixel 50.5/88.5/122.5, SCT 299/371/443/514 mm	Ref. [27] ranges '50.5<R<122.5', '299<R<514'; ATLAS JINST 3 S08003; IBL from arXiv:1803.00844 and paper Fig. 4 spacing	**REFERENCE-STATED (IBL INFERRED)**	all counts
Pixel hit = centre of hit pixel, 50x400 um (IBL 50x250 um)	Paper: 'the center of each pixel hit is used', 'pixel size rather than the resolution'; Ref. [27]: 'minimum pixel size on a sensor is 50x400 um2'	**PAPER-STATED + REFERENCE-STATED (IBL pitch INFERRED)**	all counts
Neighbouring-pixel merging	Paper: 'Hits in neightboring pixels ... are merged to a single hit' (adjacency rule INFERRED)	**PAPER-STATED**	all counts
SCT R-phi resolution 34 um	Paper: 'we conservatively assume a resolution of twice the intrinsic accuracy quoted in [27]'; Ref. [27]: '17 um (R-phi)'	**PAPER-STATED rule x REFERENCE-STATED value**	all counts
SCT z resolution 1160 um (= 2 x 580 um)	Same paper sentence (no coordinate named); Ref. [27]: 'intrinsic accuracies per module in the barrel are 17 um (R-phi) and 580 um (z)', footnote 'Arises from the 40 mrad stereo angle'	**INFERRED (paper rule applied to the reference z value; contradicted by paper Fig. 5)**	final COL1000 1/7, COL2000 3/25, EW1000 6/40, EW2000 21/94
SCT z resolution 580 um (1 x Ref. [27])	Ref. [27] value without the paper's doubling	**INFERRED (sensitivity; contradicts 'twice')**	final COL1000 3/7, COL2000 5/25, EW1000 10/40, EW2000 29/94
No SCT z measurement smearing	No source; closer to paper Fig. 5 Delta_s	**UNDOCUMENTED**	final COL1000 5/7, COL2000 11/25, EW1000 34/40, EW2000 76/94
45 mm longitudinal beam-spot smearing exists	Paper: 'a uniform, gaussian smearing with width 45 mm to the z-coordinates of all the hits'	**PAPER-STATED**	—
Beam spot = one common z shift per event, fit relative to the true vertex	Same sentence + 'We will assume that the primary vertex is identified correctly'	**INFERRED (wording ambiguous)**	SCT z 1160: COL2000 3/25, EW2000 21/94
Beam spot = one common z shift per event, fit relative to the origin	'... and is located at the origin' read literally	**INFERRED (alternative reading)**	SCT z 1160: COL2000 0/25, EW2000 3/94
Beam spot = independent 45 mm shift of every hit	'smearing ... to the z-coordinates of all the hits' read per hit	**INFERRED (alternative reading)**	fiducial COL2000 21/68, EW2000 66/186; final 0 and 0 for every SCT z (no plane found)
25 ns timing cut	Paper Sec. IV (location of the cut INFERRED)	**PAPER-STATED**	all counts
Delta_w = sqrt(lambda_2) (Eq. 11)	Paper Eq. 11; paper Fig. 4 band corresponds to 0.5 sqrt(lambda_2)	**PAPER-STATED**	no count changes
Trigger: pT(QQbar) > 200 GeV over matched events	Paper Sec. III B	**PAPER-STATED**	COL1000 101/419, COL2000 207/873, EW1000 89/511, EW2000 186/1045

Setting	Source	Status	Resulting counts
Generator settings (matching scale, PDF, scales, tune)	not stated in the paper	**UNDOCUMENTED**	trigger as above; 36 variations give EW 0.13-0.19
Pile-up: mu = 50, 45 mm per vertex, ATLAS tracking-efficiency removal, pixel merging	Paper Sec. III D; ATL-PHYS-PUB-2015-051; pool 20 000 events (paper: 125 000)	**PAPER-STATED (pool size differs)**	signal region 0/5000
Converted photon: conversion in the beam pipe, equal energy sharing, 1% probability; overlay with pile-up	Paper Sec. III D (1% derived from '~1 pb -> ~10 fb')	**PAPER-STATED**	signal region 0/9139 (z 1.16), 0/9139 (0.58), 30/9139 (none); with pile-up 0/9139

Appendix B. Paper-figure reproductions

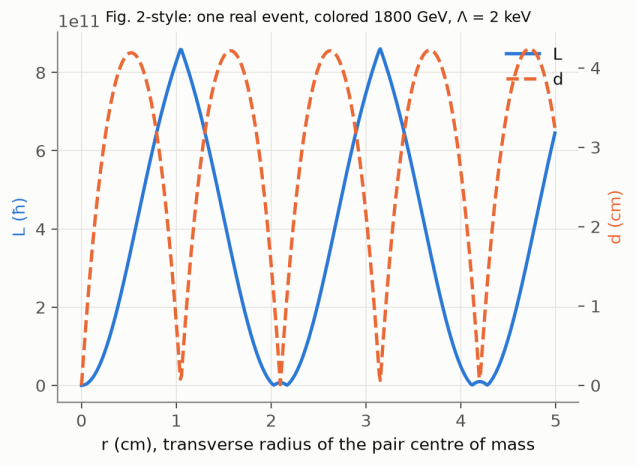


Fig. 2 style: angular momentum L and quirk separation d for one colored event (COL2000 event 107). L peaks near 9×10^{11} ħ, the paper's $\approx 10^{12}$ ħ scale.

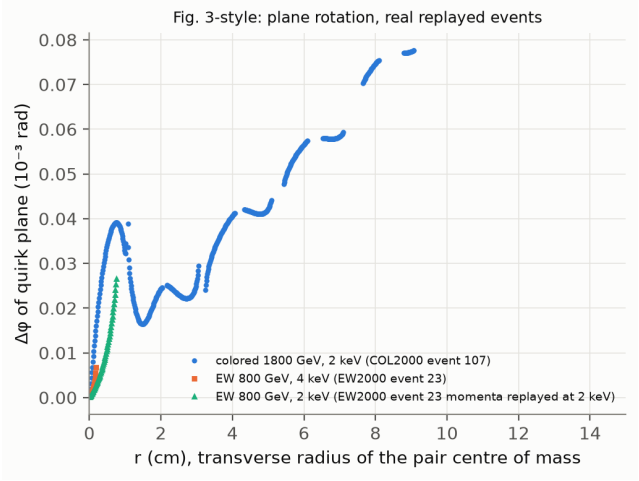


Fig. 3 style: the quirk plane rotates by less than 10^{-4} rad, so the hits stay on one plane.

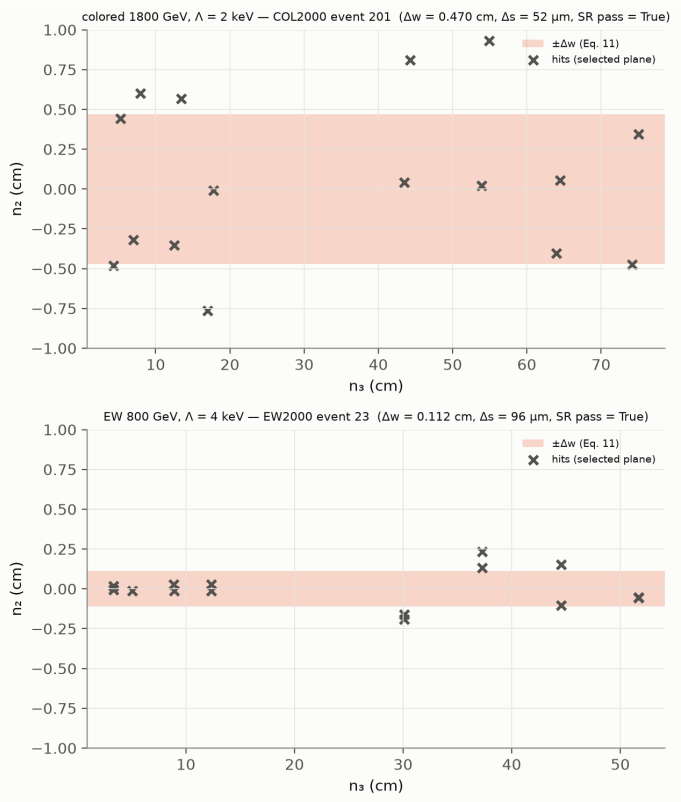


Fig. 4 style: hits of one colored (COL2000 event 201) and one EW (EW2000 event 23) event in the plane's own coordinates, SCT z 1.16 mm. Both pass the signal region (Δs = 52 and 96 μm). Band = ±Δw.